DULANA CHATHURMA

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LinkedIn | GitHub (1.2k+ Contributions • 1.1k+ Commits • 167+ ★ Stars) | Medium | Portfolio

EDUCATION

University of Kelaniya — Kelaniya, Sri Lanka
BSc (Hons) in Software Engineering — Expected Graduation: April 2029

- **Current GPA: (3.30 / 4.00)**
- **Relevant Coursework:** Data Structures and Algorithms, Object-Oriented Programming, Problem Solving, Databases, Web Development, Operating System ,Mobile Application Development

H / Rajapaksa Central College — Weeraketiya, Sri Lanka
G.C.E. Advanced Level – Physical Science Stream — (2023/2024)

- Combined Mathematics: A | Physics: B | Chemistry: B

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

- **Microsoft** Learn, **AWS** Cloud Practitioner, **Oxford** Online English, Python & Java Advanced **(Udemy)**, Programming Courses (**HackerRank**) , Programming Courses – Advanced Python & Web Design **(University of Moratuwa)**.

PROJECT EXPERIENCE

HealthPulse AI - Full-Stack Mobile & Web Companion | ***React, Node.js, Express, MongoDB*** — **Oct 2026**

- Developed a Cloud-Native Trilingual Health & Wellness Platform featuring real-time BMI tracking analytics and symptom-driven smart doctor recommendations.
- Built a secure backend with Node.js and MongoDB for persistent health vitals logging inside an interactive mobile frame.

VoxBridge AI - Real-Time Speech Interpreter | ***React, Node.js, Socket.IO, WebRTC, Gemini API*** — **Oct 2026**

- Built a low-latency speech-to-speech interpreter for live streams and meetings, powered by Google Gemini 2.5 Flash.
- Implemented dynamic VAD chunking and auto-ducking audio control using the Web Audio API and Socket.IO.

AuraTouch AI - Gesture-Controlled Desktop Navigation | ***Python, OpenCV, MediaPipe, PyAutoGUI*** — **Oct 2026**

- Developed a touchless desktop navigation and mouse control system using real-time hand & Voice gesture tracking.
- Led computer vision and gesture mapping in a team project using OpenCV, MediaPipe and PyAutoGUI.

AstroPulse-AIOps Observability Gateway | ***Python, Ballerina, React, Claude AI*** — **Sep 2026**

- Engineered a Cloud-Native Integration Gateway leveraging Ballerina and Python to streamline and monitor microservice telemetry logs in real-time.
- Integrated Claude AI to analyze system anomalies and generate automated Root Cause Analysis (RCA), significantly reducing troubleshooting time for cloud architectures.

Bare-Metal x86 Operating System | ***C, x86 Assembly, QEMU*** — **Aug 2026 – Sep 2026**

- Built a bare-metal x86 OS from scratch with a custom bootloader, round-robin scheduler and multithreading.
- Implemented PMM and heap memory management plus a custom RAM-disk file system, tested in QEMU.

Vehicle-Spare-Parts Management System | ***PHP, MySQL, JavaScript, HTML5, CSS3*** — **Aug 2026 – Sep 2026**

- Developed a full-stack E-Commerce and Management System built to streamline inventory, product cataloging, and order fulfillment processes.
- Designed a robust relational database schema using MySQL to handle real-time tracking of spare parts and manage customer transactions efficiently.

VisionScope AI | ***Python, Flask, OpenCV, MediaPipe, React.js*** — **Feb 2026 – Mar 2026**

- Developed an AI-powered real-time human feature detection application with precise face and eye tracking capabilities to ensure driver safety.
- Implemented machine learning models using OpenCV and MediaPipe for continuous webcam analysis and driver drowsiness detection.

LEADERSHIP & INVOLVEMENT

IEEE Sri Lanka Section | *Student Member* ***(ID: 102613884)*** — Aug 2026 – Present

- Participate actively in national technical workshops and networking sessions, engaging with industry professionals and adapting to emerging global tech standards.

LED KLN | *Active Member* ***(Registration No: 263092)*** — Jun 2026 – Present

- Collaborate with cross-functional student teams to execute university-wide initiatives, fostering continuous learning and innovative problem-solving.

Gavel Club, University of Kelaniya | *Coordinator* — 2025 – Present

- Coordinate weekly skill-development events and public speaking sessions for undergraduates, improving communication and leadership capabilities.

❖ **Junior Hack 7.0** | *Participant* — 2025 | University of Kelaniya

❖ **Paramount 5.0** — 2025 | Department of Mathematics, University of Kelaniya

SKILLS & INTERESTS

- **Programming Languages:** Java, Python, JavaScript, Ballerina, PHP, C, x86 Assembly
- **Frontend Technologies:** React.js, HTML5, CSS3, Tailwind CSS
- **Backend & Databases:** Node.js, Express, Flask, REST APIs, Microservices, Socket.IO, MongoDB, MySQL
- **Tools & Platforms:** Git, GitHub, CI/CD, VS Code, QEMU
- **Specialized Areas:** Artificial Intelligence, Machine Learning, Full-Stack Development, UI/UX Design, Operating Systems